Discovering Knowledge in Data

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Chapter 10
Hierarchical and *k*-Means Clustering

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Clustering Task

- Clustering refers to grouping records, observations, or tasks into classes of similar objects
- Cluster is collection records similar to one another
- Records in one cluster dissimilar to records in other clusters
- Clustering is unsupervised data mining task
- Therefore, no target variable specified
- Clustering algorithms segment records and maximize homogeneity in subgroups
- Similarity to records outside cluster minimized

Clustering Task *(cont'd)*

- For example, Nielsen PRIZM, developed by Claritas, Inc. provides demographic profiles of geographic areas, according to zip code
- PRIZM segmentation system clusters zip codes in terms of lifestyle types
- Example: Clusters identified for 90210 Beverly Hills, CA
  - Cluster 01: *Upper Crust Estates*

    "The nation's most exclusive address, Upper Crust is the wealthiest lifestyle in America, a Haven for empty-nesting couples between the ages of 45 and 64. No segment has a higher concentration of residents earning over $100,000 a year and possessing a postgraduate degree. And none has a more opulent standard of living."
  - Cluster 03: Movers and Shakers
  - Cluster 04: *Young Digerati*
  - Cluster 07: *Money and Brains*
  - Cluster 16: *Bohemian Mix*

Clustering Task *(cont'd)*

- Clustering Tasks in Business and Research
  - Target marketing for niche product, without large marketing budget
  - Accounting auditing: Segment behavior into benign and suspicious categories
  - As a dimension-reduction tool when data set has hundreds of attributes
  - Gene expression clustering, where genes exhibit similar characteristics

  - Clustering often performed as preliminary step in data mining process
  - Clustering results used as input to other data mining techniques

Clustering Task *(cont'd)*

- Applying cluster analysis to enormous databases helpful
- Reduces search space for downstream algorithms

- Cluster analysis addresses similar issues encountered in classification
  - Similarity measurement
  - Recoding categorical variables
  - Standardizing and normalizing variables
  - Number of clusters

Clustering Task *(cont'd)*

- Measuring Similarity
  - Euclidean Distance measures distance between records

$$d_{\text{Euclidean}}(\mathbf{x},\mathbf{y}) = \sqrt{\sum_i (x_i - y_i)^2}$$

where

$$\mathbf{x} = x_1, x_2, \ldots, x_m \text{ and } \mathbf{y} = y_1, y_2, \ldots, y_m$$

represent the m attribute value of two records

  - Other distance measurements include City-Block Distance and Minkowski Distance

$$d_{\text{City-Block}}(\mathbf{x},\mathbf{y}) = \sum_i |x_i - y_i|$$

$$d_{\text{Minkowski}}(\mathbf{x},\mathbf{y}) = \sum_i |x_i - y_i|^q$$

Clustering Task *(cont'd)*

- "Different From" function measures similarity between categorical attributes

$$\text{different}(x_i, y_i) = \begin{cases} 0 & \text{if } x_i = y_i \\ 1 & \text{otherwise} \end{cases}$$

- Substitute *different(x,y)* for each categorical attribute in Euclidean Distance function
- Normalizing data enhances performance of clustering algorithms
- Use Min-max Normalization or Z-Score Standardization

$$\text{Z-Score Standardization} = \frac{X - \text{mean}(X)}{\text{standard deviation}(X)}$$

$$\text{Min-Max Normalization} = \frac{X - \min(X)}{\max(X) - \min(X)}$$

Clustering Task *(cont'd)*

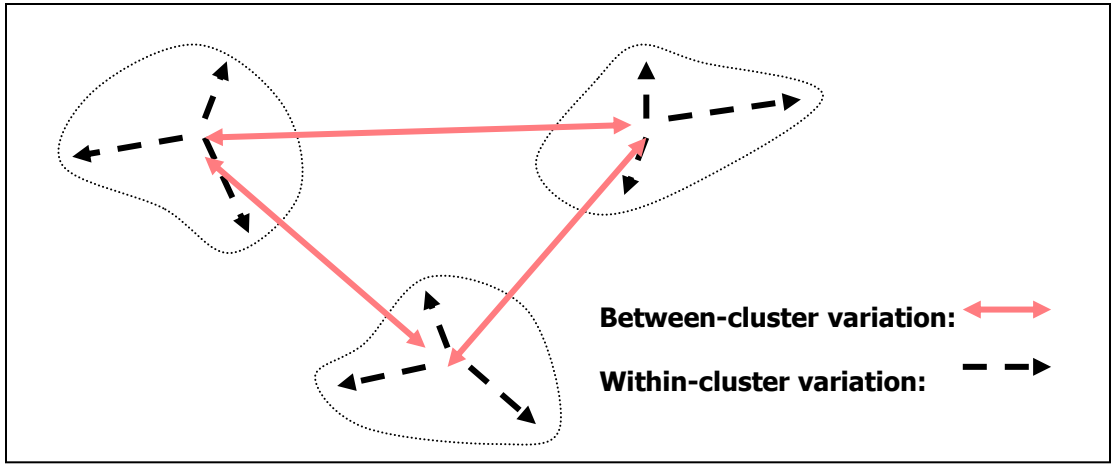


Figure 10.1

- Clustering identifies groups of highly-similar records
- Algorithms construct clusters where <u>between-cluster variation</u> (BCV) large, as compared to <u>within-cluster variation</u> (WCV)
- Analogous to concept behind analysis of variance

Hierarchical Clustering Methods

- Clustering algorithms either Hierarchical or Non-Hierarchical

- Hierarchical
  - Treelike cluster structure (dendogram) created through recursive partitioning (Divisive Methods) or combining (Agglomerative Methods) existing clusters
  - Divisive Methods
  - All records initialized into single cluster
  - At each iteration, most dissimilar record split off into separate cluster
  - Continues until each record represents single cluster

Hierarchical Clustering Methods *(cont'd)*

- Agglomerative Methods
- Each observation initialized to become own cluster
- At each iteration two closest clusters aggregated together
- Number of clusters reduced by one, each step
- Eventually, all records combined into single cluster
- Agglomerative more popular hierarchical method
- Therefore, focus remains on this approach

- Measuring distance between records straightforward once recoding and normalization applied
- However, how is distance between clusters determined?

Hierarchical Clustering Methods *(cont'd)*

- Distance Between Clusters
  - Several criteria examined to determine distance between clusters, A and B
  - Single Linkage
  - Known as Nearest-Neighbor Approach
  - Minimum distance between any record in cluster A, and any record in cluster B
  - Cluster similarity based on <u>most similar records</u> from each cluster
  - Tends to form long, slender clusters
  - Sometime heterogeneous records clustered together

Hierarchical Clustering Methods *(cont'd)*

- Complete Linkage
- Known as Farthest-Neighbor Approach
- Maximum distance between any record in cluster A, and any record in cluster B
- Cluster similarity based on <u>most dissimilar records</u> from each cluster
- Compact, sphere-like clusters formed
- All records in cluster within given diameter of other records

- Average Linkage
- Designed to reduce dependence of cluster-linkage to extreme values, such as most similar or dissimilar

Hierarchical Clustering Methods *(cont'd)*

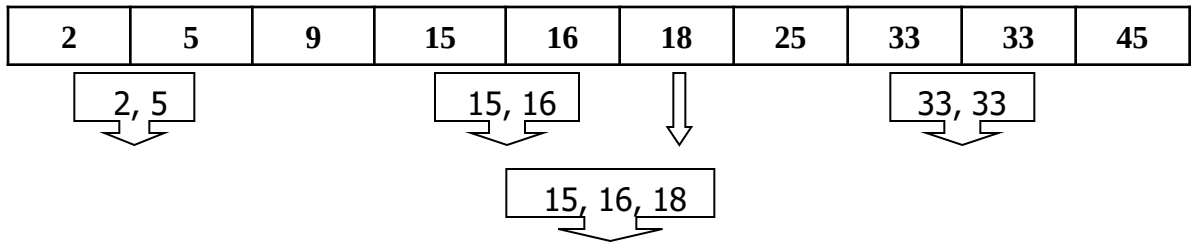
- Measure is average distance of records in cluster A, from records in cluster B
- Resulting clusters have approximately equal within-cluster variability

- Next, linkage methods examined using small data set

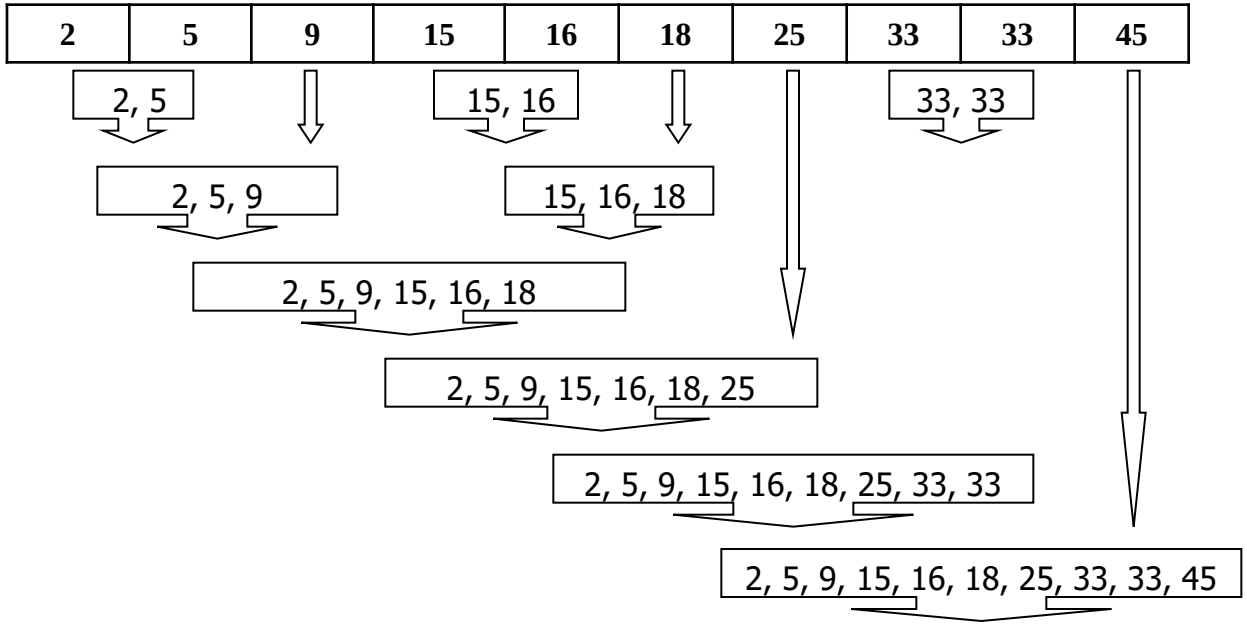
2 5 9 15 16 18 25 33 33 45

Single-Linkage Clustering

- To begin, each record assigned to own cluster
- Single-linkage seeks minimum distance between any two records, in separate clusters
- Step 1: Minimum cluster distance is between clusters {33} and {33}. Distance = 0, clusters combined
- Step 2: Clusters {15} and {16} combined, where distance = 1
- Step 3: Cluster {15, 16} combined with cluster {18}
- Step 4: Clusters {2} and {5} combined



Single-Linkage Clustering *(cont'd)*



- Agglomeration continues similarly Steps 4 – 9
- Above, last cluster {2, 5, 9, 15, 16, 18, 25, 33, 33, 45} contains all records in data set

Complete-Linkage Clustering

- Complete-linkage explored using sample data

  2 5 9 15 16 18 25 33 33 45

- Distance among records in two clusters farthest from each other minimized

- Step 1: Each cluster contains single record
  No difference between single and Complete-linkage
  Clusters {33} and {33} combined
- Step 2: Clusters {15} and {16} combined
  No difference between single and Complete-linkage

Complete-Linkage Clustering *(cont'd)*

- Step 3: Complete-linkage diverges from Single-linkage

  Farthest neighbors between {15, 16} and {18} are 15 and 18, distance = 3

  Clusters {2} and {5} also have distance = 3

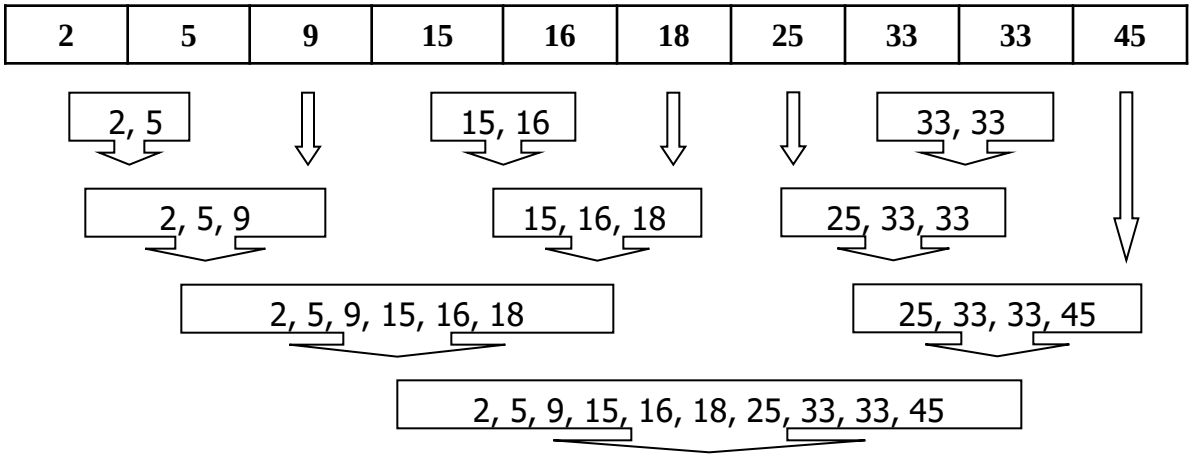
  Algorithm silent regarding ties

  Result, {2, 5} arbitrarily chosen

- Complete-linkage procedure continues for Steps 4 – 9, until all records contained in same cluster

Complete-Linkage Clustering *(cont'd)*

- Figure illustrates Complete-linkage agglomerative clustering on sample data



k-Means Clustering

- *k*-Means effective at finding clusters in data

- *k*-Means Algorithm
  - Step 1: Analyst specifies *k* = number of clusters to partition data
  - Step 2: *k* records randomly assigned to initial clusters
  - Step 3: For each record, find cluster center
    Each cluster center "owns" subset of records
    Results in *k* clusters, $C_1, C_2, \ldots, C_k$
  - Step 4: For each of *k* clusters, find cluster <u>centroid</u>
    Update cluster center location to centroid
  - Step 5: Repeats Steps 3 – 5 until convergence or termination

k-Means Clustering *(cont'd)*

- Nearest criterion in Step 3 typically Euclidean Distance

- Determining Cluster Centroid
  - Assume *n* data points $(a_1, b_1, c_1), (a_2, b_2, c_2), \ldots, (a_n, b_n, c_n)$
  - Centroid of points is center of gravity of points
  - Located at point $(\Sigma\, a_i/n, \Sigma\, b_i/n, \Sigma\, c_i/n)$

  - For example, points (1, 1, 1), (1, 2, 1), (1, 3, 1), and (2, 1, 1) have centroid

$$\left(\frac{1+1+1+2}{4}, \frac{1+2+3+1}{4}, \frac{1+1+1+1}{4}\right) = (1.25, 1.75, 1.00)$$

k-Means Clustering *(cont'd)*

- *k*-Means algorithm terminates when centroids no longer change
- For *k* clusters, $C_1, C_2, \ldots, C_k$, all records "owned" by cluster remain in cluster

- Convergence criterion may also cause termination
- For example, no significant reduction in mean squared error (MSE):

$$MSE = \frac{SSE}{N-k} = \frac{\sum_{i=1}^{k} \sum_{p \in C_i} d(p, m_i)^2}{N-k}$$

where *SSE* represents the *sum of squares error*, $p \in C_i$ represents each data point in cluster *i*, m_i represents the centroid (cluster center) of cluster *i*, *N* is the total sample size and *k* is the number of clusters

k-Means Clustering *(cont'd)*

- Clustering algorithms seek to construct clusters of records such that the between-cluster variation is large compared to the within-cluster variation
- Analogous to ANOVA, we define a *pseudo-F statistic*:

$$F_{k-1,N-k} = \frac{MSB}{MSE} = \frac{SSB/k-1}{SSE/N-k}$$

where *SSE* is defined as above, *MSB* is the *mean square between*, and *SSB* is the *sum of squares between* clusters, defined as

$$SSB = \sum_{i=1}^{k} n_i \cdot d(m_i, M)^2$$

where n_i is the number of records in cluster *i*, m_i is the centroid of cluster *i*, and *M* is the grand mean of all the data

k-Means Clustering *(cont'd)*

- MSB represents the between-cluster variation and MSE represents the within-cluster variation
- "Good" cluster has large pseudo-F statistic – where the between cluster variation is large, compared to within-cluster variation
- As K-means algorithm proceeds, MSB increases, MSE decreases, and F increases

Example of *k*-Means Clustering at Work

- Assume k = 2 to cluster following data points

a	b	c	d	e	f	g	h
(1, 3)	(3, 3)	(4, 3)	(5, 3)	(1, 2)	(4, 2)	(1, 1)	(2, 1)

- Step 1: k = 2 specifies number of clusters to partition
- Step 2: Randomly assign k = 2 cluster centers

  For example, m_1 = (1, 1) and m_2 = (2, 1)

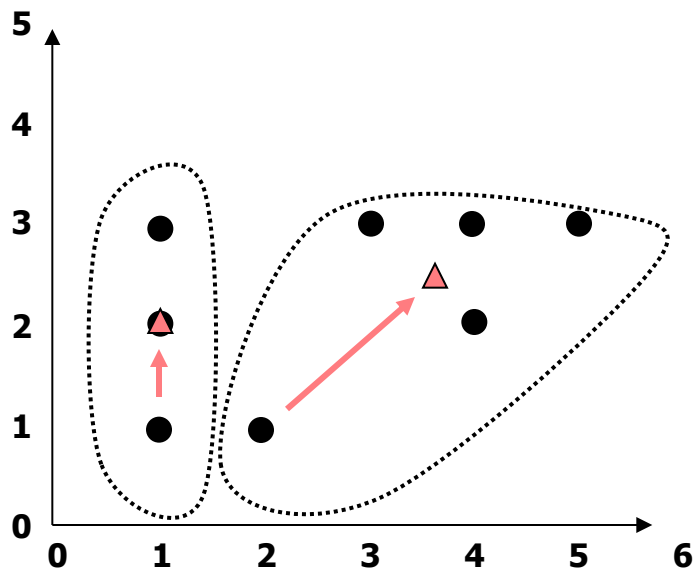
- First Iteration
  - Step 3: For each record, find nearest cluster center

    Euclidean distance from points to m_1 and m_2 shown

Point	a	b	c	d	e	f	g	h
Distance from m_1	2.00	2.83	3.61	4.47	1.00	3.16	0.00	1.00
Distance from m_2	2.24	2.24	2.83	3.61	1.41	2.24	1.00	0.00
Cluster Membership	C_1	C_2	C_2	C_2	C_1	C_2	C_1	C_2

Example of *k*-Means Clustering at Work *(cont'd)*

- Step 4: For *k* clusters, find cluster centroid, update location
- Cluster 1 = [(1 + 1 + 1)/3, (3 + 2 + 1)/3] = (1, 2), Cluster 2 = [(3 + 4 + 5 + 4 + 2)/5, (3 + 3 + 3 + 2 + 1)/5] = (3.6, 2.4)
- Figure shows movement of clusters m*1* and m*2* (triangles) after first iteration of algorithm



Example of *k*-Means Clustering at Work *(cont'd)*

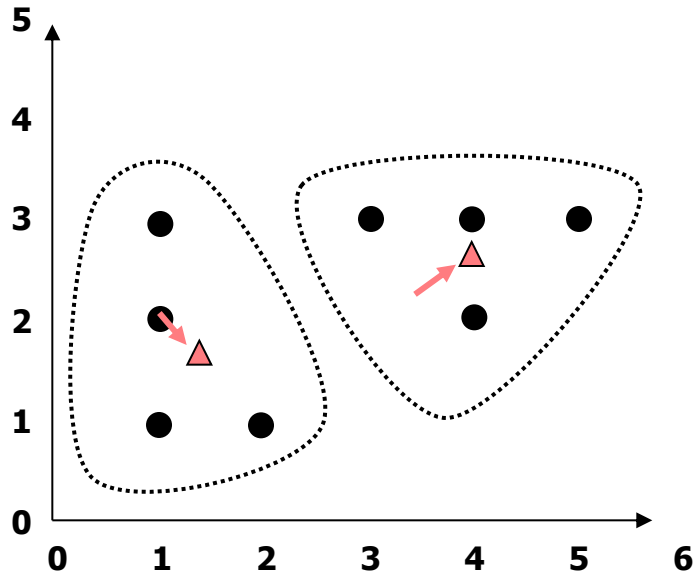
- Step 5: Repeats Steps 3 – 4 until convergence or termination

- Second Iteration
  - Repeat procedure for Steps 3 – 4
  - Again, for each record find nearest cluster center m_1 = (1, 2) or m_2 = (3.6, 2.4)
  - Table below shows how *h* moved to cluster 1

Point	a	b	c	d	e	f	g	h
Distance from m_1	2.00	2.83	3.61	4.47	1.00	3.16	0.00	1.00
Distance from m_2	2.24	2.24	2.83	3.61	1.41	2.24	1.00	0.00
Cluster Membership	C_1	C_2	C_2	C_2	C_1	C_2	C_1	C_1

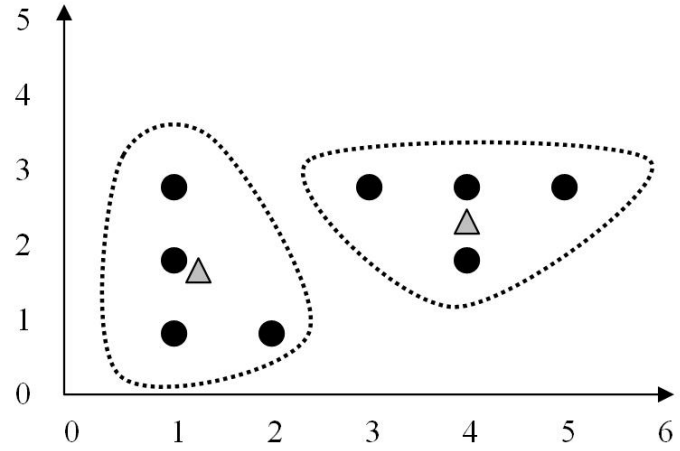
Example of *k*-Means Clustering at Work *(cont'd)*

- Cluster centroids updated to m_1 = (1.25, 1.75) or m_2 = (4, 2.75)
- After Second Iteration, cluster centroids shown to move slightly



Example of *k*-Means Clustering at Work *(cont'd)*

- Third (Final) Iteration
  - Repeat procedure for Steps 3 – 4
  - Now, for each record find nearest cluster center m_1 = (1.25, 1.75) or m_2 = (4, 2.75)
  - This time, no records shift cluster membership
  - Centroids remain unchanged, therefore algorithm terminates



Behavior of MSB, MSE and pseudo-F as the k-means algorithm proceeds

- First pass

$$SSB = \sum_{i=1}^{k} n_i \cdot d(m_i, M)^2 = 3 \cdot d\big((1,2),(2.625,2.25)\big)^2 + 5 \cdot d\big((3.6,2.4),(2.625,2.25)\big)^2 = 12.975$$

$$MSB = \frac{SSB}{k-1} = \frac{12.975}{2-1} = \mathbf{12.975}$$

$$SSE = \sum_{i=1}^{k} \sum_{p \in C_i} d(p, m_i)^2 = 2^2 + 2.24^2 + 2.83^2 + 3.61^2 + 1^2 + 2.24^2 + 0^2 + 0^2 = 36$$

$$MSE = \frac{SSE}{N-k} = \frac{36}{6} = \mathbf{6}$$

$$F = \frac{MSB}{MSE} = \frac{12.975}{6} = \mathbf{2.1625}$$

- Expectation is for MSB to increase, MSE to decrease and F to increase in subsequent passes
- Results obtained are
  - Second pass: MSB = 17.125, MSE = 1.313333, F = 13.03934
  - Third pass: MSB = 17.125, MSE = 1.041667, F = 16.44
- We have achieved maximum between-cluster variation (as measured by MSB), compared to within-cluster variation (as measured by MSE)

Example of *k*-Means Clustering at Work *(cont'd)*

- Summary
  - *k*-Means not guaranteed to find to find global maximum pseudo-F
  - Instead, local minimum found
  - Invoking algorithm using variety of initial cluster centers improves probability of achieving global maximum
  - One approach places first cluster at random point, with remaining clusters placed far from previous centers (Moore)

  - What is appropriate value for *k*?
  - Potential problem for applying *k*-Means
  - Analyst may have *a priori* knowledge of *k*

Example of *k*-Means Clustering at Work *(cont'd)*

- Outer loop to algorithm possible
- Cycles through different *k* values
- Results compared, selecting solution with largest F statistic
- Alternatively, other clustering algorithms like BIRCH can select optimal number of clusters

- What attributes to use as input?
- Some attributes likely more relevant than others
- Apply axis-stretching methods for quantifying attribute relevance discussed in Chapter 7
- In Chapter 11 we examine Kohonen networks, related to artificial neural networks

Application of *k*-Means Clustering Using SAS Enterprise Miner

- *k*-Means algorithm in SAS Enterprise Miner analyzes *churn* data set
- Uses FASTCLUS procedure, a version of *k*-Means
- Data set contains 20 attributes with 3,333 records
- Target variable describes if customer left company ("churned")
- 9 attributes passed to algorithm, with $k = 3$
- Categorical
- International Plan, Voice Main Plan
- Numeric (values Normalized)
- Account Length, Voice Mail Messages, Day Minutes, Evening Minutes, Night Minutes, International Minutes, Customer Service Calls
  - Min-Max Normalization was applied to all numerical variables

Application of *k*-Means Clustering Using SAS Enterprise Miner *(cont'd)*

- Three clusters uncovered vary significantly in size
- Cluster 1: 92 records, Cluster 2: 2,411 records, and Cluster 3: 830 records
- Cluster Profiling helps understand types of records comprising clusters

Application of *k*-Means Clustering Using SAS Enterprise Miner *(cont'd)*

- Clustering results window from Enterprise Miner shows pie chart profile of *International Plan* across cluster membership
  - Cluster 1: All members adopted International Plan
  - Cluster 2: Fraction adopted plan
  - Cluster 3: None chose plan

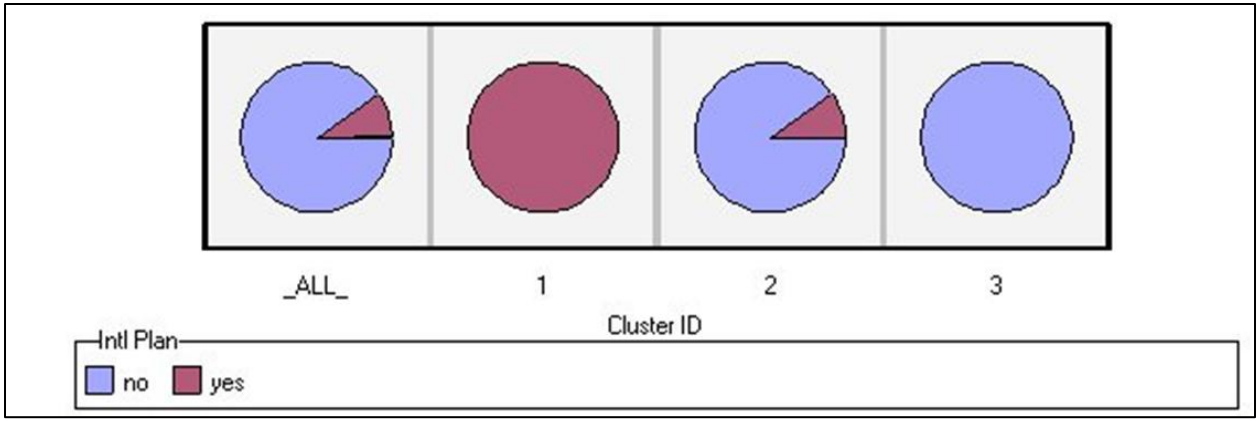


Figure 10.7

Application of *k*-Means Clustering Using SAS Enterprise Miner *(cont'd)*

- Below, results show profile of *Voice Mail Plan* across cluster membership
- Clusters 1 and 3: All members adopted Voice Mail Plan
- Cluster 2: None chose plan

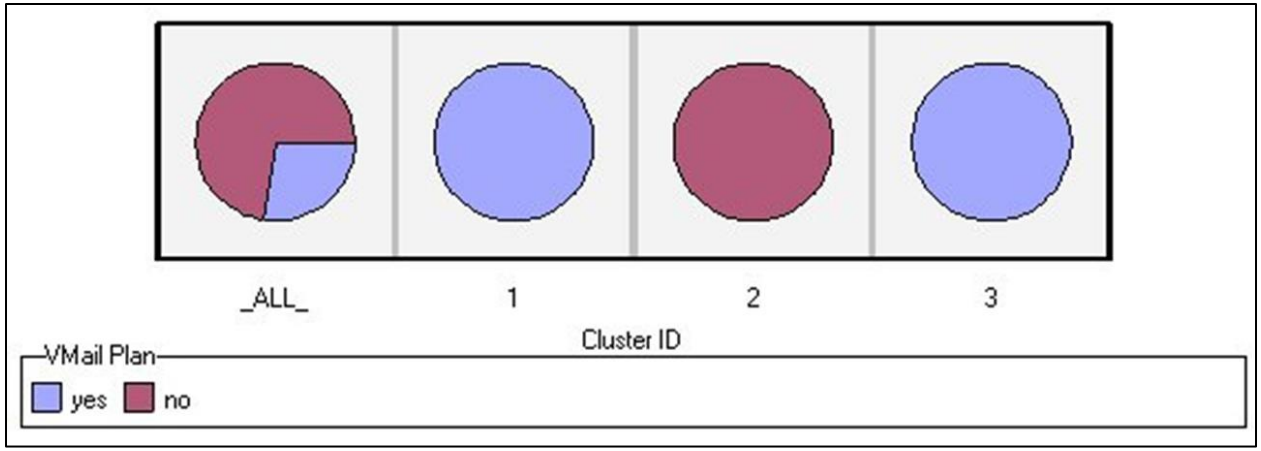


Figure 10.8

Application of *k*-Means Clustering Using SAS Enterprise Miner *(cont'd)*

- *k*-Means discriminates records according to *Voice Mail Plan* field value
- Divides data set perfectly among adopters and non-adopters
- Algorithm relies heavily on categorical attributes
- Mean values for numerical variables shows little variation across clusters (see table below)

Table 10.5

Cluster	Freq.	AcctLength_m	VMailMessage	DayMins_mm
1	92	0.4340639598	0.5826939471	0.5360015616
2	2411	0.4131940041	0	0.5126334451
3	830	0.4120730857	0.5731159934	0.5093940185
Cluster	**EveMins_mm**	**NightMins_mm**	**IntMins_mm**	**CustServCalls**
1	0.5669029659	0.4764366069	0.5467934783	0.1630434783
2	0.5507417372	0.4773586813	0.5119784322	0.1752615328
3	0.5564095259	0.4795138596	0.5076626506	0.1701472557

Application of *k*-Means Clustering Using SAS Enterprise Miner *(cont'd)*

- Cluster Summary
  - Cluster 1: Sophisticated Users
    - Smallest group of customers
    - Adopted International and Voice Mail Plans
  - Cluster 2: The Average Majority
    - Largest segment of customer base
    - Some have adopted Voice Mail Plan
    - None have adopted International Plan
  - Cluster 3: Voice Mail Users
    - Medium-sized group
    - All have chosen Voice Mail Plan
    - None have adopted International Plan

Using Cluster Membership to Predict Churn

- Suppose clusters used to predict "churn" for classification
- Proportion for those churning analyzed for each cluster
- Diagram shows that proportion of churners higher for those adopting International Plan
- Company should investigate why those adopting plan leaving company at higher rate

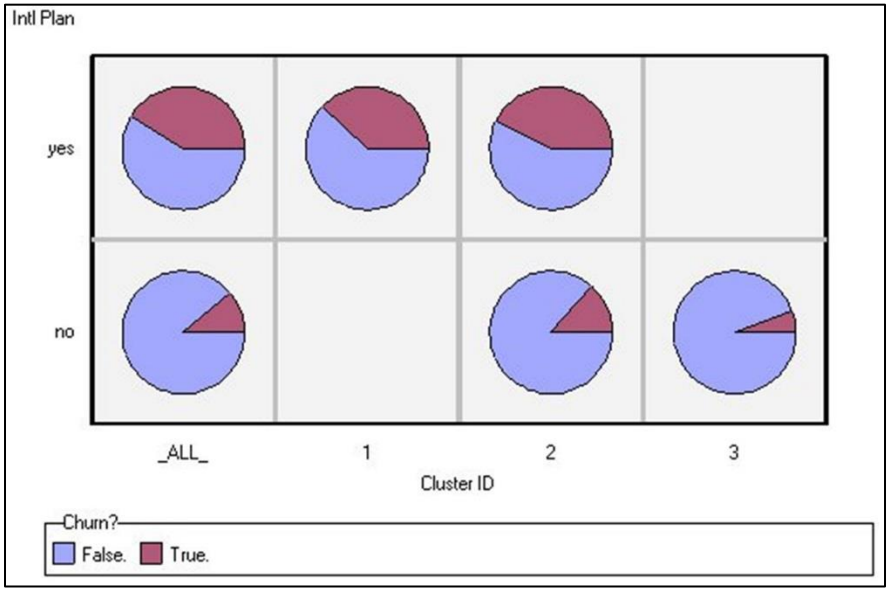


Figure 10.10

Using Cluster Membership to Predict Churn *(cont'd)*

- Recall proportion of churners lower, for those participating in Voice Mail Plan
- Figure confirms lower churn rate for customers in Cluster 3 (all adopters of Voice Mail Plan)

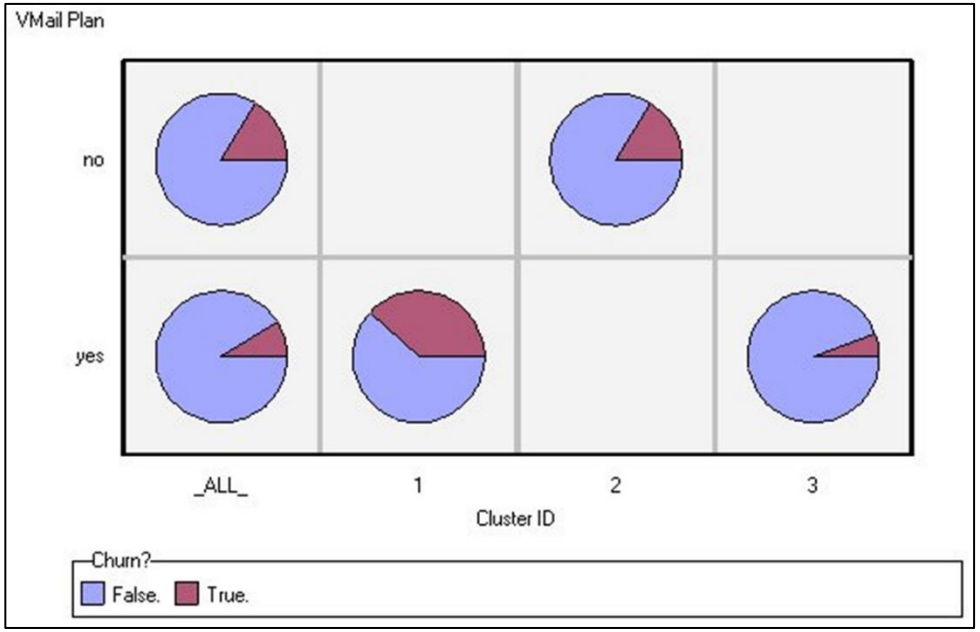


Figure 10.11